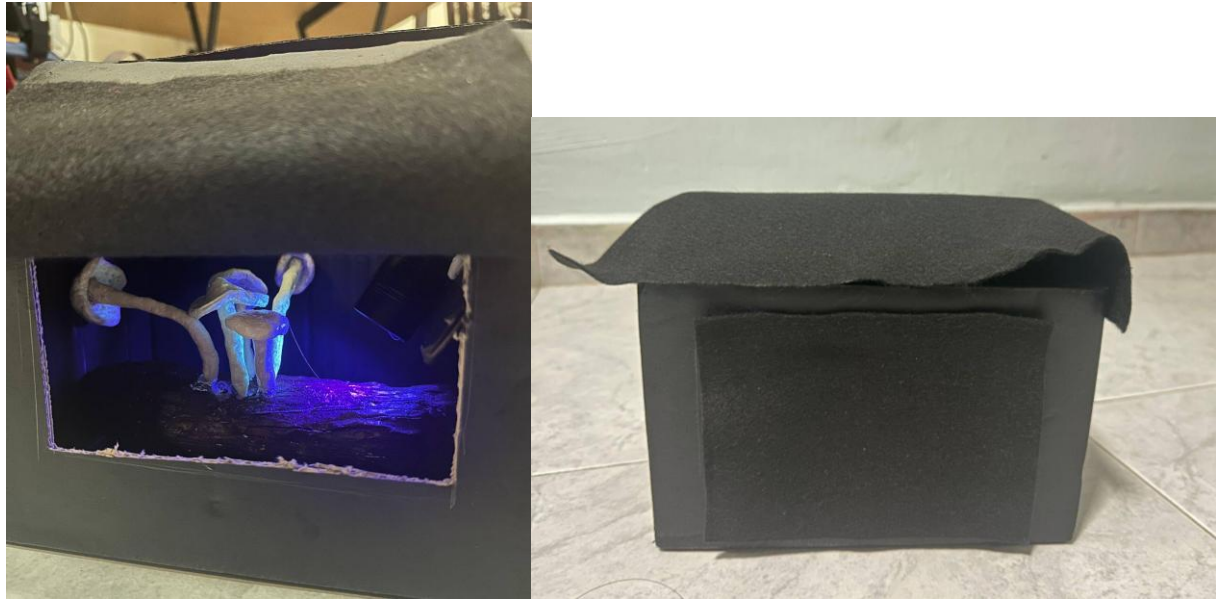


Name: Vinny Ong



The physical item that I made is a handmade clay model of *Filoboletus manipularis*, a bioluminescence pore fungus found in Singapore. The model shows a cluster of five mushrooms growing on a decayed log and enclosed in a black cardboard box to imitate the night environment and showcase the luminescence.

Filoboletus manipularis (Berk.) Singer is a saprotrophic bioluminescent fungus (Kassim et al., 2020) that is widely distributed on decayed wood throughout Asia and the Pacific tropical region (Vydryakova et al., 2014). It was previously spotted in Singapore in 1954 but was only officially confirmed in 2009, suggesting a rediscovery (Lee, 2022).

Morphological Diversity

Filoboletus Manipularis has a diverse range of morphology, ranging from the shape, size, colour and luminosity (Vydryakova et al., 2014), all of which I have represented in the clay model. Even though mushrooms that are grown in the same cluster should potentially exhibit similar characteristics (Vydryakova et al., 2014), I varied the sizes and colours to demonstrate the different morphological characteristics that would occur in nature. Besides the above-mentioned variation, the pileus of *Filoboletus Manipularis* also varies from conical to rounded or plane to depressed shape (Vydryakova et al., 2014). The colours of the mushroom at maturity also vary, where mature specimens exhibit white to cream or pale pink colours and those that are still developing would display brownish specks around the pileus (Vydryakova et al., 2014). Despite the diverse morphological variations, all the specimens still belonged to the same species, indicating that *F. manipularis* species exist with high genetic plasticity.

Alternatively, *F. manipularis* have angular poroid hymenophores ranging from 0.5 to 1mm wide (Vydryakova et al., 2014, Lee, 2022) instead of the usual gilled structures that are found in many fungi in the Mycenaceae family. To ensure that the pores are represented as accurately as possible, its texture and openings were hand-sculpted using toothpicks and sculpting tools.

Bioluminescence Mechanism

To mimic the actual bioluminescence that the fungi would produce in nature, I painted the mushroom with glow-in-the-dark paint. Bioluminescence is a biochemical process where living organisms emit light and during this process, luciferase enzymes will oxidise luciferins substrate in the presence of oxygen, releasing energy in the form of visible light (Ke & Tsai, 2022). The light falls within the wavelength of 530 nm, which is the yellow-green range (Ke & Tsai, 2022). Therefore, *F. manipularis* emits a green bioluminescence light. Additionally, there is variation in the bioluminescence produced by *F. manipularis*, where some are present around the pileus, pores, stipe and entire body and others do not show any at all (Vydryakova et al., 2014). To better understand the variation, I have shown it in the model.

Ecological Significance

Currently, there are only 125 named species of bioluminescence fungi, with most of them dominating the *Mycena* genus (Lu et al., 2024), making them a rare group of fungi as compared to the others. All of them are Basidiomycota except for one (Lu et al., 2024). The ecological function of bioluminescence in fungi is still widely studied, with many hypothesizing that its purpose could be to attract insects or other organisms to aid in spore dispersal (Stevani et al., 2024, Lu et al., 2024) or as a protection against other nocturnal fungivores (Lu et al., 2024). However, there are certain species including *F. manipularis* itself that have bioluminescence emitting from the stipe or mycelium, which challenges the hypothesis regarding the use of insects for spore dispersal. Therefore, it increases the complexity of the studies regarding bioluminescence in fungi (Lu et al., 2024), making it interesting and worthwhile to study about.

The study of bioluminescent fungi like *F. manipularis* provides an opportunity to explore new possibilities in the understanding of fungal biodiversity and through the study of fundamental genetic mechanisms involved in light production from living organisms, we will have a better understanding about the fungal ecosystem and their ecological roles.

References

- Kassim, M. H. S. B., Wasti, I. G., Subari, I. A. A. M. H., Tang, T. A. G., P. L., Chong, C. C., Subramaniam, N., & Seelan, J. S. S. (2020). Macrofungi of Sungai Kangkawat Research Station, Imbak Canyon Conservation Area, Sabah, Malaysia. *The Malayan Nature Journal*, 72(3), 371–388.
- Ke, H.-M., & Tsai, I. J. (2022). Understanding and using fungal bioluminescence – Recent progress and future perspectives. *Current Opinion in Green and Sustainable Chemistry*, 33(100570). <https://doi.org/10.1016/j.cogsc.2021.100570>
- Lee, S. M. L. (2022). The status of bioluminescent fungal species in Singapore. *NATURE IN SINGAPORE Supplement 1*. <https://doi.org/10.26107/NIS-2022-0124>
- Lu, W., Priyashantha, A. K. H., Galappaththi, M. C. A., Tibpromma, S., Dai, D.-Q., Patabendige, N. M., Premarathne, B. M., Kulasuriya, D. M., Ediriweera, A. N., Nimalrathna, T., Suwannarach, N., Lumyong, S., Tang, A., Shao, S.-C., & Karunarathna, S. C. (2024). Fungal Bioluminescence: Past, Present, and Future. *Diversity*, 16(9), 539. <https://doi.org/10.3390/d16090539>
- Stevani, C. V., Zamuner, C. K., Bastos, E. L., Nóbrega, B. B. de, Soares, D. M. M., Oliveira, A. G., Bechara, E. J. H., Shakhova, E. S., Sarkisyan, K. S., Yampolsky, I. V., & Kaskova, Z. M. (2024). The living light from fungi. *Journal of Photochemistry and Photobiology C: Photochemistry Reviews*, 58(100654). <https://doi-org.libproxy1.nus.edu.sg/10.1016/j.jphotochemrev.2024.100654>
- Vydryakova, G. A., Morozova, O. V., Redhead, S. A., & Bissett, J. (2014). Observations on morphologic and genetic diversity in populations of *Filoboletus manipularis* (Fungi: Mycenaceae) in southern Viet Nam. *Mycology*, 5(2), 81–97. <https://doi.org/10.1080/21501203.2014.902402>